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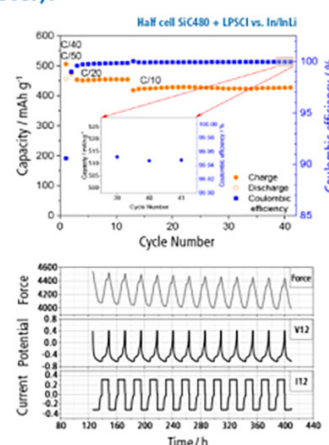
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Formal Equivalence between Doyle–Fuller–Newman and Dual-Continuum Lithium-Ion Battery Models

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Upscaled electrochemical models are widely used to simulate lithium-ion batteries efficiently, with the Doyle–Fuller–Newman (DFN, or P2D) model serving as the standard framework. Recently, dual-continuum (DC) models have emerged as an alternative—these avoid simplifying assumptions on particle geometry through a closure problem that is solved on the true microstructure. On the surface, the DFN and DC models appear starkly different. However, in this work we demonstrate that they are *formally* equivalent when: (i) active-material particles are isolated spheres of uniform radius, (ii) the “transient” DC formulation is used, (iii) initial concentration is uniform, and (iv) diffusivity does not depend on radial position. A similar equivalence exists between single-particle and single-continuum models. This equivalence provides fundamental insight into both approaches: the DC model reduces exactly to the DFN for idealised spherical geometries while enabling simulation of arbitrarily complex microstructures through closure problems. We validate this equivalence through simulations and discuss implications for developing efficient battery models, particularly regarding the time convolution describing memory effects in diffusion processes.

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Detailed models of lithium-ion batteries can be built that resolve transport equations directly on the battery microstructure.^{1–6} Due to the large computational cost of this approach, upscaled models of lithium-ion batteries have become extremely popular. Amongst such upscaled approaches, the Doyle–Fuller–Newman (DFN) model has emerged as the cornerstone (also known as the pseudo-2D, or P2D model).⁷ Since its inception, several simplifications (e.g. the single particle model⁸) and extensions (e.g. the multi-particle DFN model⁹) of the DFN formulation have become popular – Brosa-Planella et al.¹ published a detailed review on the range of upscaled models that exist.

One family of upscaled formulations was omitted from this review: dual-continuum (DC) models.^{10–13} This is because they have rarely appeared in the battery modelling literature until recently, despite their widespread use in other applications (e.g., heat-transfer¹⁴ and petroleum engineering^{15–17}). DFN- and DC-type models (which are abbreviated further to just DFN and DC throughout this article) can be derived using upscaling theories, such as multi-scale asymptotics^{10,18–21} or volume averaging.^{22,23} Both the DFN and DC upscale the conservation equations for mass and charge in the electrolyte phase, and that for charge within the active material (AM). However, they take a fundamentally different approach to modelling mass transport in the AM particles—this is illustrated in Fig. 1. The DFN employs a *hybrid* approach, with the aforementioned macroscale/homogenised equations being coupled to a micro-scale description of mass transport within the AM particles. By contrast, DC models are *fully* homogenised, and thus have an upscaled equation for mass transport in the AM. The challenge with hybrid models, such as the DFN, is their computational cost—if the real geometry of the AM particles is kept, not much advantage is gained compared to the detailed microscale model discussed above. It is for this reason that the DFN makes the strong assumption that particles can be treated as isolated spheres. The challenge with DC models, on the other hand, is estimating AM surface concentration, which is critical in calculating reaction rate. Since their upscaled equation for AM concentration describes a volume-averaged value, a mapping must be used to relate this to a surface-averaged concentration.

We recently derived a DC model using the volume-averaging technique,^{12,13} which overcame this challenge by using a closure variable to relate volume-averaged and surface-averaged concentrations. This closure variable is defined by a transport problem solved on the real electrode microstructure. Crucially, our model required no assumption on the AM particle shape. We then extended this derivation to the “transient” DC model, which included the time dependency of the closure variable, rather than its steady-state value used in the original derivation. One of the hallmarks of this model is the appearance of a time convolution in the governing equations. Although not previously applied to lithium-ion batteries, transient closure problems are classical in the volume-averaging literature.^{24–26} Their improved accuracy (compared to a steady-state closure) is particularly noticeable in systems where the closure variable varies over a timescale similar to the characteristic timescales in the upscaled model. In the context of batteries, the transient behavior of the closure variable was found to be most important when AM diffusion is slow, and under changing battery loads (application of a current to a battery at rest state, for example). We also simplified the DC model to a “single-continuum” (SC) model, which we compared to the popular single-particle model (SPM) family.

The DFN and DC models appear to be starkly different. In this article, we show that despite their apparent differences, if the geometrical assumption employed by the two models is the same, they are *formally* equivalent. Likewise, the transient SC and SPM models are also equivalent under such conditions. More broadly, equivalence exists between the DC model and a *general* hybrid model (see Fig. 1), regardless of the electrode geometry. We begin by demonstrating equivalence theoretically, and then complement the theory with simulations; the latter serves as validation of the developments made, and demonstrates the DC model’s advantage when departing from spherical geometries. This equivalence offers insight into the fundamental nature of the models, highlights the strengths of each approach, and inspires research into new methods to efficiently solve battery models.

Model Summary

Definitions.—A schematic of an electrode is shown in Fig. 2a. We define the electrolyte domain as Ω_e , whilst that of the AM is Ω_k ,

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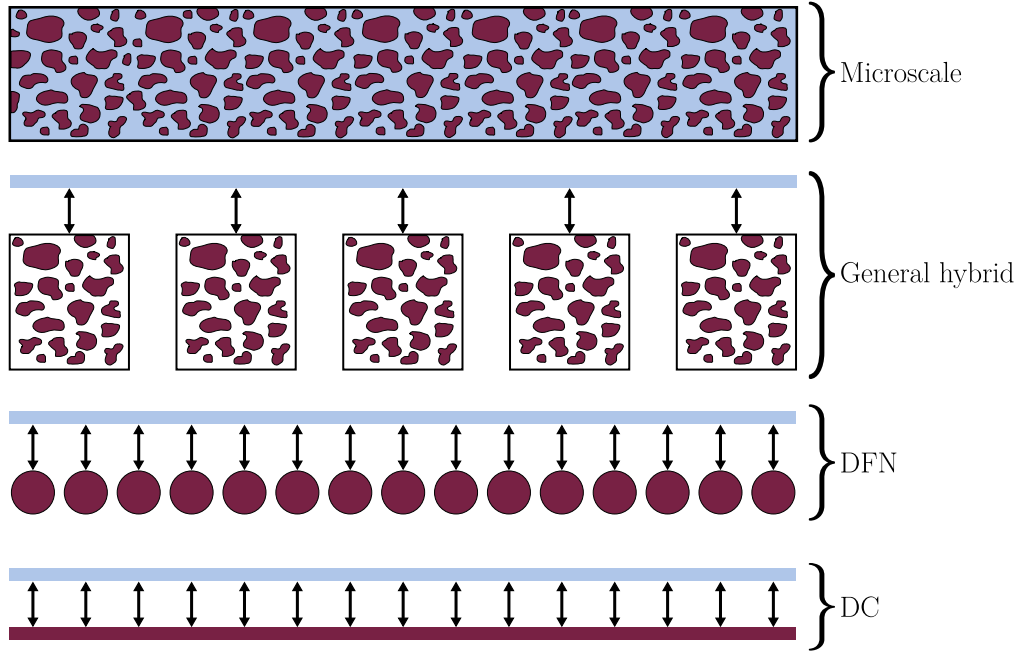


Figure 1. Top to bottom: the microscale model resolves the full electrode geometry; the general hybrid model couples a macroscale electrolyte description to an auxiliary AM microstructural cell at each macroscopic point; the DFN replaces the cell with an isolated sphere; the DC model upscales the active-material phase. Electrolyte is blue and AM is burgundy.

where $k \in \{n, p\}$ to denote the negative or positive electrode respectively. The AM-electrolyte interface is defined as $\partial\Omega_{k,e}$, and the domain encompassing both phases is called $\Omega = \Omega_e \cup \Omega_k$. Further, Ω_k can be decomposed into subdomains for different AM particles, Ω_{k_i} , where i is the particle index (see Fig. 2b). The boundary between the electrolyte and Particle i is $\partial\Omega_{k_i,e}$, and $\partial\Omega_{k_i,k_j}$ is the boundary between Particle i and its neighbour, Particle j . Spatial position within the system is denoted by $\mathbf{x} \in \Omega$, whilst time is $t \in \mathbb{R}$ (discussed below). An averaging set centred at position $\mathbf{x}$ is defined as $\mathcal{V}(\mathbf{x}) \subset \Omega$, and the definition of the volume-averaging operator is given by,

$$\langle u \rangle^\gamma(\mathbf{x}, t) \equiv \frac{1}{V_\gamma(\mathbf{x})} \int_{\xi \in \mathcal{V}_\gamma(\mathbf{x})} u(\xi, t) dV_\xi, \quad [1]$$

where $\mathcal{V}_\gamma(\mathbf{x}) \subset \mathcal{V}(\mathbf{x})$ is the subset of phase γ within the averaging set and $V_\gamma(\mathbf{x}) \equiv \int_{\xi \in \mathcal{V}_\gamma(\mathbf{x})} dV_\xi$. Note that the general definition of an averaging operator employs a spatial convolution.^{27–29}

The DFN model.—In DFN’s hybrid approach, the microscale PDEs describing electrolyte mass/charge conservation and AM charge conservation are all upscaled. Therefore, these PDEs operate on volume-averaged values of electrolyte concentration, electrolyte potential, and AM potential: $\langle c_e \rangle^e$, $\langle \phi_e \rangle^e$, and $\langle \phi_k \rangle^k$ respectively. Since these quantities are volume averaged, they are defined at *all* points within Ω (provided the averaging volume is chosen such that $V_e(\mathbf{x}) \neq \emptyset$ and $V_k(\mathbf{x}) \neq \emptyset$). The DFN does not upscale the PDE describing AM concentration, c_k , but rather solves the microscale diffusion equation within auxiliary particle domains. This introduces a new spatial variable: in the standard version of the DFN, AM particles are assumed to be isolated spheres, and thus this variable is $r \in [0, R]$, where R is the particle radius. Whilst Fig. 1 is helpful for understanding model differences, a potentially misleading feature is that it shows a finite number of particles for the DFN – this is only true for the discretised form of the model. The analytical form of the DFN, on the other hand, possesses a *continuum* of particles, and every point $\mathbf{x}$ is coupled to a particle domain, as illustrated in Fig. 2b.

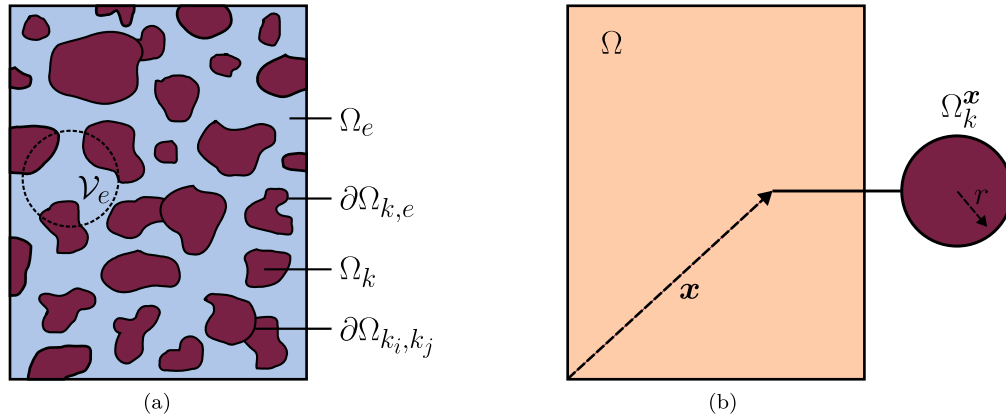


Figure 2. (a) A schematic of an electrode, with the AM phase (Ω_k), the electrolyte (Ω_e), the boundary between electrolyte and AM ($\partial\Omega_{k,e}$), a boundary between two AM particles ($\partial\Omega_{k_i,k_j}$), and an example averaging set ($\mathcal{V}_e$). (b) A schematic of DFN’s key assumptions. A new color is assigned to Ω , to highlight that it does not identify phases, but simply the set containing all $\mathbf{x}$ within $\Omega_e \cup \Omega_k$. The auxiliary particle domain at any point $\mathbf{x}$ is defined as $\Omega_k^{\mathbf{x}}$.

As mentioned above, three of the four conservation equations are identical for the DFN and DC, and thus the reader is referred to our previous work for their definition.¹² Here, we focus on the defining difference between the two models—mass conservation in the AM particles. In the standard DFN formulation, mass flux (N_k) is modelled with Fickian diffusion, and reaction rate (j_k^d) is modelled by Butler-Volmer (BV) kinetics.^{1,7} Further, j_k^d is assumed to be invariant over the particle surface, thus giving a spherically symmetrical initial-boundary value problem (IBVP),

$$\partial_t c_k(\mathbf{x}, r, t) = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_k) \text{ for } 0 \leq r \leq R, \quad [2a]$$

$$N_k = H(t) \frac{j_k^d}{F} \text{ at } r = R, \quad [2b]$$

$$N_k = 0 \text{ at } r = 0, \quad [2c]$$

$$c_k = c_0 \text{ for } t \leq 0, \quad [2d]$$

where c_k is lithium concentration in the AM particles, R is the sphere radius, c_0 is the initial AM concentration, $\mathbf{x}$ is location within Ω , and r is position within the auxiliary particle domain. Notice that c_k has been defined for $t \in \mathbb{R}$ by employing a Heaviside function, $H(t)$. This allows us to benefit from certain convolution properties later in our developments. The flux and BV expressions are,

$$N_k = -D_k \frac{\partial c_k}{\partial r}, \quad [3a]$$

$$j_k^d = 2j_{k0}^d \sinh\left(\frac{F}{2R_u T} \eta_k^d\right), \quad [3b]$$

in which

$$j_{k0}^d = Fk_k \sqrt{c_k|_R \langle c_e \rangle^e (c_{k,\max} - c_k|_R)}, \quad [4a]$$

$$\eta_k^d = \langle \phi_k \rangle^k - \langle \phi_e \rangle^e - U_k(c_k|_R). \quad [4b]$$

The electrolyte diffusivity, conductivity, and transport number are D_e , σ_e , t_+^0 (respectively). Finally, F is Faraday's constant, R_u is the universal gas constant, T is temperature, j_{k0}^d is the exchange current density, η_k^d is the reaction overpotential, U_k is the open-circuit potential, k_k is the rate constant, D_k is the AM diffusivity, and $c_{k,\max}$ is the maximum concentration of lithium in the AM.

The transient DC model.—The transient DC model,¹³ herein denoted DC^{1t} for consistency with the original publication, upscales the AM mass balance, and thus describes volume-averaged concentration, $\langle c_k \rangle^k$, rather than microscale concentration, c_k . As for the DFN, variables in DC^{1t} will be defined over $t \in \mathbb{R}$ by employing a Heaviside function, to leverage convolution properties later on,

$$\partial_t \langle c_k \rangle^k(\mathbf{x}, t) = -H(t) \frac{A}{FV_k} j_k^{1t}, \text{ in } \Omega, \quad [5a]$$

$$\langle c_k \rangle^k = c_0 \text{ for } t \leq 0, \quad [5b]$$

where

$$A = \int_{\partial\Omega_{k,e}} dA, \quad [6a]$$

$$V_k = \int_{\Omega_k} dV, \quad [6b]$$

$$j_k^{1t} = 2j_{k0}^{1t} \sinh\left(\frac{F}{2R_u T} \eta_k^{1t}\right), \quad [6c]$$

and

$$j_{k0}^{1t} = Fk_k \sqrt{\langle c_k \rangle_A^k \langle c_e \rangle^e (c_{k,\max} - \langle c_k \rangle_A^k)}, \quad [7a]$$

$$\eta_k^{1t} = \langle \phi_k \rangle^k - \langle \phi_e \rangle^e - U_k(\langle c_k \rangle_A^k). \quad [7b]$$

Note that the only difference between the rate expressions for the DFN and DC^{1t} models, j_k^d and j_k^{1t} (Eqs. 3b and 6c), is the AM surface concentration input, $c_k|_R$ or $\langle c_k \rangle_A^k$ respectively. As in the original publication, the latter is defined as,

$$\langle c_k \rangle_A^k = \langle c_k \rangle^k + \langle s^t \rangle_A \star \partial_t j_k^{1t}, \quad [8]$$

where

$$\langle s^t \rangle_A = \frac{1}{A} \sum_{i=1}^n \int_{\partial\Omega_{k_i,e}} s_i^t dA, \quad [9]$$

in which s_i^t is the closure variable and n is the number of particles in the averaging set. Note that the original work used A_i in the place of $\partial\Omega_{k_i,e}$, but that these are interchangeable. The time convolution $f \star g$ is defined as

$$f \star g = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau, \quad [10]$$

and s_i^t is given by the closure problem solution for Particle i ,

$$\partial_t s_i^t = D_k \Delta s_i^t + H(t) \frac{1}{F} \frac{A_i}{V_{k_i}} \text{ in } \Omega_{k_i}, \quad [11a]$$

$$-\mathbf{n} \cdot D_k \nabla s_i^t = H(t) \frac{1}{F} \text{ at } \partial\Omega_{k_i,e}, \quad [11b]$$

$$\mathbf{n} \cdot \nabla s_i^t = \mathbf{n} \cdot \nabla s_j^t \text{ at } \partial\Omega_{k_i,k_j}, \quad [11c]$$

$$s_i^t = s_j^t \text{ at } \partial\Omega_{k_i,k_j}, \quad [11d]$$

$$s_i^t = 0 \text{ for } t \leq 0, \quad [11e]$$

where $\mathbf{n}$ is the normal vector pointing outwards from the AM domain, and

$$A_i = \int_{\partial\Omega_{k_i,e}} dA, \quad [12a]$$

$$V_{k_i} = \int_{\Omega_{k_i}} dV. \quad [12b]$$

Note that Eq. 11 is solved directly on the microstructure (such as in Fig. 2a), thus avoiding any assumption on particle shape. We highlight two important assumptions that are made in the derivation of DC^{1t}, but not in that of the DFN model. Firstly, DC^{1t} assumes that initial concentration is uniform in the particles: $c_k = c_0$ in Ω_k at $t = 0$, where c_0 is a constant value. Although such a condition is standard in

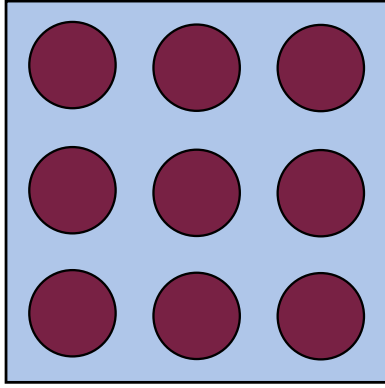


Figure 3. A cross section of an electrode built from isolated spheres with uniform radii (burgundy), immersed in an effective phase (blue) possessing both ion and electron conductivity.

battery simulations, the DFN does not assume this, and it would be feasible to have $c_0(\mathbf{x}, r)$. An extension of DC^{1t} could nonetheless be developed which handles non-uniform initial conditions (ICs), but the closure problem would then be dependent on the specific IC enforced.

Secondly, to obtain the closure problem in the form shown here (Eq. 11), we assume that D_k does not vary significantly over the scale of an averaging volume. The DFN, on the other hand, allows for diffusivity varying with radial position within a particle. Note that DC models *can* handle non-linear diffusivity functions, as shown in the original publications. Precisely, the assumption needed to demonstrate equivalence between DC^{1t} and the DFN is that D_k is not a function of radial position in the latter. Equivalence will therefore still hold when diffusivity is a function of *volume-averaged* concentration, $D_k(\langle c_k \rangle^k)$. As before, an extension of DC^{1t} could be developed which considers variations in $D_k(c_k)$ across the scale of an averaging volume, but this would introduce dependency on c_k into the closure problem. Finally, DC^{1t} could also be recast using a phase splitting approach,^{30–32} which may ease the handling of more general ICs and diffusion coefficients.

Equivalence for an Electrode of Isolated Spheres

In this section, we consider an electrode made of isolated spheres, as shown in Fig. 3. In this case, no error is introduced by the DFN's assumption on particle shape. We first write DC^{1t} for this geometry. Then, we reformulate the DFN model using a time convolution to show that it is formally equivalent to DC^{1t} .

DC^{1t} for an electrode of isolated spheres.— DC^{1t} was written above in its general form. However, for an electrode of isolated spheres with uniform radii, several simplifications can be made. Firstly, since all particles are identical, the closure problem need only be solved for one particle, which will represent the entire electrode. Secondly, particles are isolated, and thus no interparticle boundaries ($\partial\Omega_{k_i,k_j}$) exist. Thirdly, since the IC is uniform, and the BC is uniform across the surface of the sphere, the problem is spherically symmetric: this facilitates simplification of the divergence operator, and also the addition of a “no-flux” regularity condition at $r = 0$. The resulting closure problem is thus,

$$\partial_t s^t = D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial s^t}{\partial r} \right) + H(t) \frac{1}{F} \frac{A}{V_k} \text{ for } 0 \leq r \leq R, \quad [13a]$$

$$-D_k \frac{\partial s^t}{\partial r} = H(t) \frac{1}{F} \text{ at } r = R, \quad [13b]$$

$$-D_k \frac{\partial s^t}{\partial r} = 0 \text{ at } r = 0, \quad [13c]$$

$$s^t = 0 \text{ for } t \leq 0. \quad [13d]$$

The spherical symmetry also means that the surface-average of the closure variable, $\langle s^t \rangle_A$ (Eq. 9), can simply be evaluated as

$$\langle s^t \rangle_A = s^t|_R, \quad [14]$$

and thus the surface-averaged concentration definition (Eq. 8) becomes

$$\langle c_k \rangle_A^{1t} = \langle c_k \rangle^k + s^t|_R \star \partial_{tj_k}^{1t}. \quad [15]$$

Although our primary interest is surface concentration, note that the microscale concentration at any point within the sphere can be reconstructed via the closure problem,

$$c_k = \langle c_k \rangle^k + s^t \star \partial_{tj_k}^{1t}. \quad [16]$$

Reformulating the DFN using the divergence theorem.—Note that in the standard form of the DFN's flux expression (Eq. 3a), the assumption of radial symmetry has already been introduced. To demonstrate the equivalence between DFN and DC^{1t} for this electrode, we start by writing the flux in a more general way,

$$\mathbf{N} = (N_r, N_\theta, N_\varphi), \quad [17]$$

where θ and φ are the polar and azimuthal angles in a spherical coordinate system, and $N_r = N_k$ (Eq. 3a). We make use of the divergence theorem,

$$\int_{\mathcal{V}_s} (\nabla \cdot \mathbf{N}) dV = \int_{\mathcal{A}_s} (\mathbf{n} \cdot \mathbf{N}) dA, \quad [18]$$

where $\mathcal{V}_s$ is the set of all points within the sphere volume, and $\mathcal{A}_s$ is the set of all points on the sphere surface. For the case where $N_\theta, N_\varphi = 0$, which is assumed by the DFN,

$$\nabla \cdot \mathbf{N} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_r), \quad [19a]$$

$$\mathbf{n} \cdot \mathbf{N} = N_r. \quad [19b]$$

Inserting this into the divergence theorem gives

$$\int_{\mathcal{V}_s} \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_r) dV = \int_{\mathcal{A}_s} N_r dA, \quad [20]$$

then exploiting the BC in Eq. 2b and noting that j_k^d does not vary across $\mathcal{A}_s$,

$$\int_{\mathcal{V}_s} \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_r) dV = H(t) \frac{1}{F} j_k^d A_s. \quad [21]$$

where $A_s = \int_{\mathcal{A}_s} dA$.

Volume-averaged form.—To manipulate the DFN model into a form similar to that of DC^{1t} , we seek a volume-averaged version of the PDE describing c_k . Returning to Eq. 2a, we integrate over $\mathcal{V}_s$,

$$\int_{\mathcal{V}_s} \partial_t c_k dV = - \int_{\mathcal{V}_s} \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_k) dV. \quad [22]$$

Because every point $\mathbf{x}$ is coupled to a single auxiliary particle domain $\Omega_k^x = \mathcal{V}_s(\mathbf{x})$, the AM phase-intrinsic average $\langle c_k \rangle^k(\mathbf{x}, t)$ (Eq. 1) reduces to an average over a single sphere,

$$\langle c_k \rangle^k = \frac{1}{V_s} \int_{V_s} c_k dV, \quad [23]$$

where $V_s = \int_{V_s} dV$. Inserting Eq. 21 into Eq. 22, and using the definition in Eq. 23 gives

$$\partial_t \langle c_k \rangle^k = -H(t) \frac{1}{F} \frac{A_s}{V_s} j_k^d. \quad [24]$$

Since the electrode is composed exclusively of isolated spheres of uniform radius, the ratio $\frac{A_s}{V_s}$ is equivalent to the global surface area to volume ratio in the electrode,

$$\frac{A_s}{V_s} = \frac{A}{V_k}, \quad [25]$$

and thus Eq. 24 can be written as

$$\partial_t \langle c_k \rangle^k = -H(t) \frac{1}{F} \frac{A}{V_k} j_k^d. \quad [26]$$

Thus far, we have begun from the DFN model and derived a PDE for $\langle c_k \rangle^k$ which takes a very similar form to that in DC¹⁷ (Eq. 5) – the only difference is the BV expression used, j_k^d or j_k^{17} .

The deviation equation.—Now, we define a deviation between the microscale concentration $c_k(\mathbf{x}, r, t)$ and the volume averaged concentration $\langle c_k \rangle^k(\mathbf{x}, t)$,

$$\tilde{c}_k = c_k - \langle c_k \rangle^k. \quad [27]$$

Since $\langle c_k \rangle^k$ is not a function of r , taking the partial derivative of Eq. 27 with respect to r gives,

$$\frac{\partial c_k}{\partial r} = \frac{\partial \tilde{c}_k}{\partial r}. \quad [28]$$

To obtain a PDE on $\tilde{c}_k(\mathbf{x}, r, t)$, we subtract Eq. 26 from Eq. 2a. The BCs and IC for this problem are found by inserting the decomposition for c_k into Eqs. 2b to 2d. The resulting IBVP is

$$\partial_t \tilde{c}_k = D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \tilde{c}_k}{\partial r} \right) + H(t) \frac{1}{F} \frac{A}{V_k} j_k^d \text{ for } 0 \leq r \leq R, \quad [29a]$$

$$-D_k \frac{\partial \tilde{c}_k}{\partial r} = H(t) \frac{j_k^d}{F} \text{ at } r = R, \quad [29b]$$

$$-D_k \frac{\partial \tilde{c}_k}{\partial r} = 0 \text{ at } r = 0, \quad [29c]$$

$$\tilde{c}_k = 0 \text{ for } t \leq 0. \quad [29d]$$

Note that the IC has been simplified since initial concentration is uniform in the particles, and thus $\langle c_k \rangle^k = c_0$ at $t = 0$. Also, D_k has been removed from the derivative terms on the basis that it is not a function of radial position, as discussed above.

Seeking a solution using a convolution.—We now present a new function, f , and will subsequently show that f is the unique solution to $\tilde{c}_k$ in Eq. 29:

$$f(\mathbf{x}, r, t) = s^d \star \partial_{tj_k^d}, \quad [30]$$

where

$$\partial_t s^d = D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial s^d}{\partial r} \right) + H(t) \frac{1}{F} \frac{A}{V_k} \text{ for } 0 \leq r \leq R, \quad [31a]$$

$$-D_k \frac{\partial s^d}{\partial r} = H(t) \frac{1}{F} \text{ at } r = R, \quad [31b]$$

$$-D_k \frac{\partial s^d}{\partial r} = 0 \text{ at } r = 0, \quad [31c]$$

$$s^d = 0 \text{ for } t \leq 0. \quad [31d]$$

Equation 30 may be interpreted as an application of Duhamel's principle.³³ The IBVP on s^d captures the particle's response to a unit Heaviside flux excitation. Since the IBVP on $\tilde{c}_k$ is linear and its spatial operator is time independent, its solution can be expressed as a time convolution of the system's impulse response, s^d , with the forcing term, j_k^d . Thus, the form of Eq. 30 is not ad hoc, but rather the natural consequence of linear superposition for parabolic IBVPs. Convolution the PDE and BCs (Eqs. 31a–31c) with $\partial_{tj_k^d}$ gives,

$$\begin{aligned} \partial_t s^d \star \partial_{tj_k^d} &= \left[D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial s^d}{\partial r} \right) \right] \star \partial_{tj_k^d} \\ &+ \frac{1}{F} \frac{A}{V_k} H(t) \star \partial_{tj_k^d} \text{ for } 0 \leq r \leq R, \end{aligned} \quad [32a]$$

$$-D_k \frac{\partial s^d}{\partial r} \star \partial_{tj_k^d} = \frac{1}{F} H(t) \star \partial_{tj_k^d} \text{ at } r = R, \quad [32b]$$

$$-D_k \frac{\partial s^d}{\partial r} \star \partial_{tj_k^d} = 0 \text{ at } r = 0. \quad [32c]$$

Now, exploiting the algebraic properties of convolutions to move derivatives (e.g. $\partial_t s^d \star j_k^d = s^d \star \partial_{tj_k^d}$), and using the fact that $\partial_t H(t) \star g(t) = g(t)$,

$$\begin{aligned} \partial_t (\partial_t s^d \star j_k^d) &= D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial}{\partial r} (\partial_t s^d \star j_k^d) \right] \\ &+ \frac{1}{F} \frac{A}{V_k} j_k^d \text{ for } 0 \leq r \leq R, \end{aligned} \quad [33a]$$

$$-D_k \frac{\partial}{\partial r} (\partial_t s^d \star j_k^d) = \frac{j_k^d}{F} \text{ at } r = R, \quad [33b]$$

$$-D_k \frac{\partial}{\partial r} (\partial_t s^d \star j_k^d) = 0 \text{ at } r = 0. \quad [33c]$$

Since $s^d = 0$ and $j_k^d = 0$ when $t \leq 0$, we can also write,

$$\partial_t s^d \star j_k^d = 0 \text{ for } t \leq 0. \quad [34]$$

Finally, inserting the definition of $f(\mathbf{x}, r, t)$ into Eqs. 33 and 34 gives,

$$\partial_t f = D_k \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + H(t) \frac{1}{F} \frac{A}{V_k} j_k^d \text{ for } 0 \leq r \leq R, \quad [35a]$$

$$-D_k \frac{\partial f}{\partial r} = H(t) \frac{j_k^d}{F} \quad \text{at } r = R, \quad [35b]$$

$$-D_k \frac{\partial f}{\partial r} = 0 \quad \text{at } r = 0, \quad [35c]$$

$$f = 0 \quad \text{for } t \leq 0. \quad [35d]$$

Notice that the Heaviside function has been reintroduced to Eq. 35: since $j_k^d = 0$ when $t \leq 0$, $j_k^d = H(t)j_k^d$. This has been done to facilitate comparison with Eq. 29.

Equivalence.—Comparing equations 29 and 35 reveals that they take identical form. These equations are both linear, parabolic IBVPs with inhomogeneous Neumann BCs. Under the condition that $j_k^d(t) \in L^2(0, T)$, where T is the final integration time, classical PDE theory guarantees a unique weak solution to such problems.³³ Therefore, f (Eq. 30) gives the unique solution for $\tilde{c}_k(\mathbf{x}, r, t)$,

$$f(\mathbf{x}, r, t) = \tilde{c}_k(\mathbf{x}, r, t). \quad [36]$$

Now, we seek an expression for the surface concentration, which is required to calculate $j_k^d(t)$. Using Eqs. 27, 30, and 36, an expression for the surface concentration may be constructed,

$$c_k|_R = \langle c_k \rangle^k + s^d|_R \star \partial_{tj_k^d}. \quad [37]$$

Comparing Eqs. 37 and 15 reveals that the surface-concentration expressions for the DFN and DC models are identical,

$$c_k|_R = \langle c_k \rangle_A^{lt}. \quad [38]$$

Crucially, the BV expressions in the two models, j_k^d and j_k^{lt} , only differ by the surface concentration used, $c_k|_R$ and $\langle c_k \rangle_A^{lt}$ respectively. Therefore,

$$j_k^d = j_k^{lt}. \quad [39]$$

We now see that j_k^{lt} and j_k^d are equivalent for this case, and thus the PDEs on $\langle c_k \rangle^k$ are also equivalent (Eqs. 5 and 26). Finally, since all other PDEs within the two models are identical, the two models are formally equivalent under the following conditions: electrode AM particles are isolated spheres of uniform radii, initial concentration is uniform in the particles, and AM diffusivity is not a function of radial position. This equivalence is summarised in Fig. 4.

Analogy with SPM and SC Models

Whilst equivalence has been demonstrated for the DFN and DC^{lt}, the exact same arguments hold when comparing SPM-type models and the transient SC model (SC^{lt}). The terminology “SPM-type” is used since a range of SPMs exist, which differ in their treatment of the electrolyte equations.¹ However, all of these use the same approach for modelling the AM phase. They make the assumption that a single particle can represent the entire electrode, rather than a continuum of particles as in the DFN, and solve an equation very similar to Eq. 2,

$$\partial_t c_k = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_k) \quad \text{for } 0 \leq r \leq R, \quad [40a]$$

$$N_k = H(t) \frac{\bar{j}_k}{F} \quad \text{at } r = R, \quad [40b]$$

$$N_k = 0 \quad \text{at } r = 0, \quad [40c]$$

$$c_k = c_0 \quad \text{for } t \leq 0. \quad [40d]$$

The sole difference between Eqs. 40 and 2 is the calculation of reaction rate, j_k^d or $\bar{j}_k$. Whilst that in the DFN is coupled to the electrolyte equations via BV kinetics (Eq. 3b), that in the SPM is decoupled from the electrolyte phase, due to the assumption that reaction is evenly distributed across the electrode,

$$\bar{j}_k = \lambda_k \frac{i_{\text{app}}}{L_k a_k}, \quad [41]$$

where $\lambda_n = -1$, $\lambda_p = 1$, i_{app} is the applied current density, L_k is the length of the electrode, and a_k is the specific surface area of the electrode. In this work, we will focus on the SPM_e model presented by Marquis et al.⁸ (where “e” symbolises the inclusion of electrolyte dynamics).

Turning our attention to the SC^{lt} model, this reduces DC^{lt} by assuming that $\langle c_k \rangle^k$ does not vary significantly along the thickness of the electrode, x . Similarly to the SPM, reaction is assumed to be evenly distributed. This results in a single ordinary differential equation to describe the AM in each electrode, since $\langle c_k \rangle^k(\mathbf{x}, t)$ becomes $\langle c_k \rangle^k(t)$,

$$\partial_t \langle c_k \rangle^k(t) = -H(t) \frac{A}{FV_k} \bar{j}_k(t). \quad [42]$$

The expression for surface concentration in this model, $\langle c_k \rangle_A^{\text{sc}lt}$, is,

$$\langle c_k \rangle_A^{\text{sc}lt} = \langle c_k \rangle^k + \langle s^t \rangle_A \star \partial_{tj_k^d}. \quad [43]$$

Besides the treatment of the AM phase, all remaining equations in the SPM_e and SC^{lt} models are identical.¹³ Since the DFN and SPM_e models for the AM phase are so similar in form (Eqs. 40 vs. 2), as are the DC^{lt} and SC^{lt} models (Eqs. 5 vs. 42), it is clear that the arguments advanced above for DFN-DC equivalence will also be applicable for the SPM_e and SC^{lt} models. For brevity, the comparison of SPM_e and SC^{lt} will therefore not be developed further.

Simulations

To complement the demonstration above, simulations are presented for all models, ensuring that predictions from the DFN and DC^{lt} (or SPM_e and SC^{lt}) are indeed equivalent. These serve as a validation of the developments made, and also demonstrate the advantages of DC^{lt} (or SC^{lt}) when departing from spherical geometries.

Methods.—All models are parameterised and solved for two half cells: the first is a nickel manganese cobalt oxide (NMC) electrode, and the second is graphite. For this application, we prefer to simulate half cells to isolate the behavior of each electrode. All electrode properties have been taken from Chen’s work,³⁴ in which a commercial cell (LGM 50) was deconstructed and analysed, presenting a full parameter set for the DFN; values are shown in Table I. The remaining parameters required to complete the model are given in Table II. A 1C discharge is simulated. Chen reported stoichiometries at 100% and 0% state of charge (SOC)—these values are used to define the initial AM concentration and also the point at which the discharge terminates (respectively). The closure variable for the DC^{lt} and SC^{lt} models was solved using COMSOL Multiphysics,³⁵ and all remaining methods and numerical information required for solution are exactly as defined in our previous work.^{12,13}

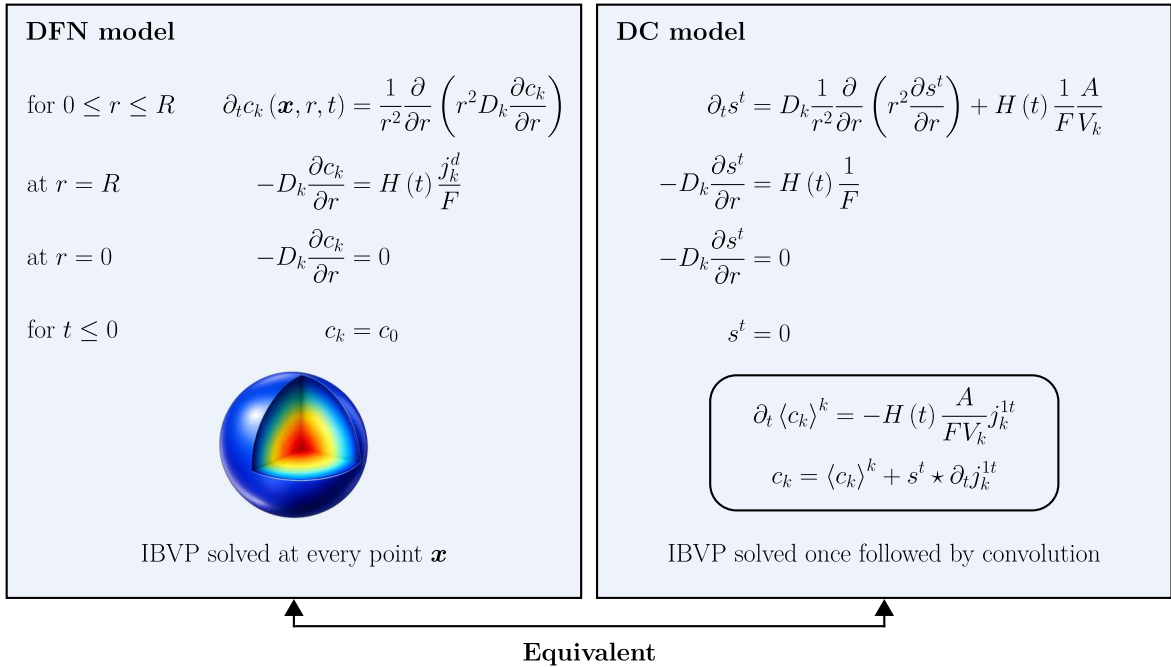


Figure 4. Summarising the equivalence between DFN and DC^{1t} models.

Table I. The AM material properties for each electrode, all taken from Chen's work.³⁴

Property name	Symbol	Negative electrode	Positive electrode
Stoichiometry at 0% SOC	$x_{k,0}$	0.0279	0.9084
Stoichiometry at 100% SOC	$x_{k,100}$	0.9014	0.2700
Maximum AM concentration [molm ⁻³]	$c_{k,max}$	33133	63104
Particle radius [μm]	R_k	5.86	5.2
Electrode length [μm]	L_k	85.2	75.6
AM volume fraction	ε_k	0.75	0.665
Porosity	ε_e	0.25	0.335
Electrolyte tortuosity	τ_e	2.0	1.73
AM conductivity	σ_k	215	0.18
AM diffusivity [m ² s ⁻¹]	D_k	3.3×10^{-14}	4.0×10^{-15}
Rate constant [Am ⁻² (m ³ mol ⁻¹) ^{1.5}]	Fk_k	6.48×10^{-7}	3.42×10^{-6}

Table II. Properties for the electrolyte, separator, and lithium foil.

Property Name	Symbol	Value	References
Thermodynamic factor	$(1 + \delta_e)$	1	36
Transference number	t_+^0	0.4	36
Electrolyte conductivity [Sm ⁻¹]	σ_e	1	36
Electrolyte diffusivity [m ² s ⁻¹]	D_e	1×10^{-10}	36
Initial electrolyte concentration [molm ⁻³]	$c_{e,0}$	1000	34
Temperature [K]	T	298.15	n/a
Separator tortuosity	$\tau_{e,s}$	2.28	37
Separator porosity	$\varepsilon_{e,s}$	0.4	37
Exchange-current density for lithium foil [Am ⁻²]	$j_{li,0}$	10	23

Closure variable results.—The solutions to the closure variable at the particle surface as a function of time, $\langle s^t \rangle_A$, are shown in Fig. 5. For the NMC particle, $\langle s^t \rangle_A$ takes longer to reach steady state due to the lower diffusivity value used. The steady-state field of s^t for the NMC particle is also given in Fig. 6.

Cell voltage results.—The cell voltage results for all simulations are shown in Fig. 7. The results show perfect agreement between the DFN and DC^{1t} models for both the NMC and graphite half cells. Likewise, the predictions of the SPMe and SC^{1t} models are identical. As an aside, the graphite half cell reaches a high voltage (~2 V) which would not be encountered in a real cell due to voltage constraints being enforced. Nonetheless, this does not affect the model comparison, but on the contrary explores a more extreme range of operating conditions than those encountered in a real cell. These simulations are not intended to serve as proof that the DC^{1t} and DFN models (or SC^{1t} and SPM) are strictly equivalent—rather, that is the role of the mathematical development presented above. Instead, they primarily function as a reassurance that nothing has been overlooked in our analysis, and that equivalent results are indeed produced.

Discussion

Geometrical assumptions.—This equivalence provides fundamental insight into the nature of the DC^{1t} model, and in particular into its advantages over the DFN. In the idealised case of an electrode composed of isolated spherical particles of uniform radius, the DC^{1t} model reduces *exactly* to the DFN. However, unlike the DFN, the DC^{1t} formulation is not restricted to the assumption of spherical particle geometry – the closure problem (Eq. 11) can be solved on *any* geometry, regardless of its complexity. The results in Fig. 8 (repeated from our previous work^{12,13}) demonstrate this point. For the electrode here, which was built from a real tomographic image, the DFN's isolated-sphere assumption introduces considerable error when compared to the direct numerical solutions (DNS) of a detailed microscale model (see the original publication for DNS details). On the other hand, DC^{1t} reproduces the DNS with high fidelity, particularly in the relaxation period of the simulation. To

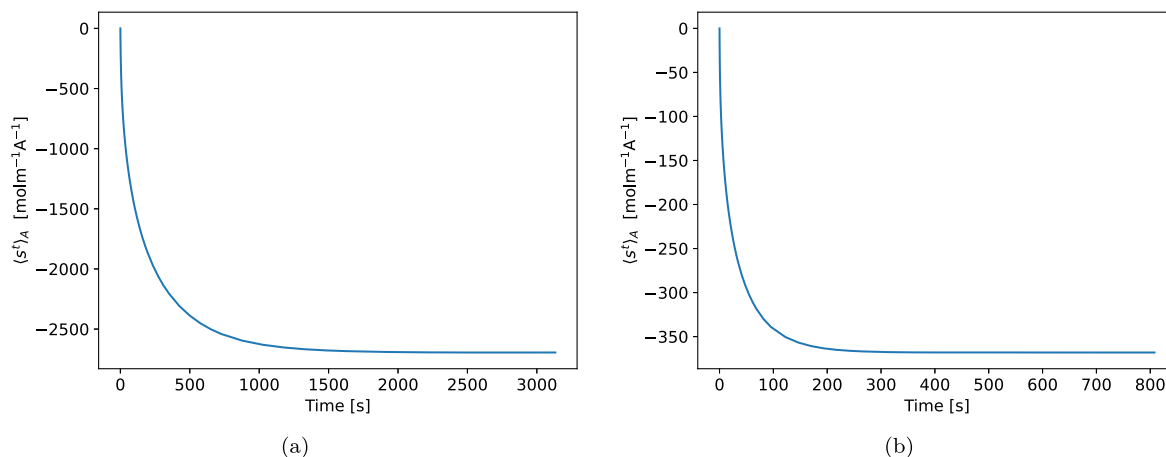


Figure 5. The surface value of the closure variable solution, $\langle s^t \rangle_A$, for the NMC (a) and graphite (b) particles.

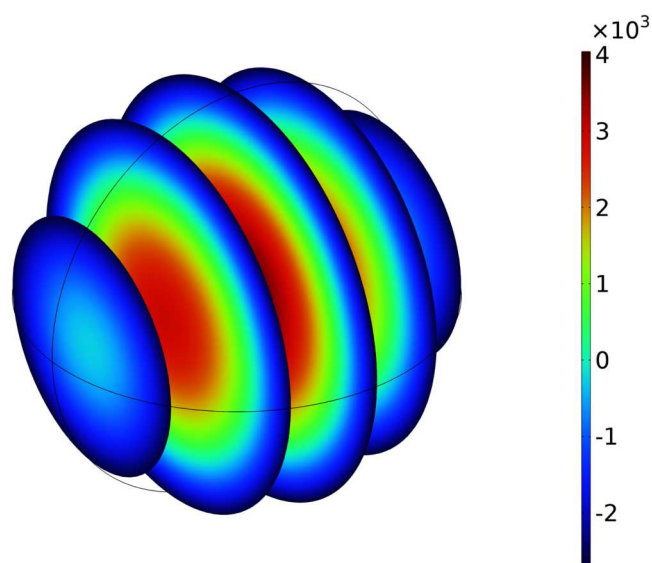


Figure 6. A slice view of the closure variable solution, s^t [molm⁻¹A⁻¹], at steady state for the NMC particle.

summarise, DC^{1t} captures AM transport characteristics identically to the DFN for spherical particles, but is not limited to modelling spherical particles. Once again, this is analogous for the comparison of SC^{1t} and SPMe.

Solving the time convolution.—Another interpretation of the contrast between the two models is that they employ different approaches to solve the time convolution. Although not explicit in its traditional form, this equivalence demonstrates that the DFN (or SPMe) essentially solves a convolution (see Eq. 37). It does so via its solution of the diffusion equation, which is executed for every simulation of the electrode, and at every point x . On the other hand, DC^{1t} (or SC^{1t}) solves the diffusion equation *once*, which characterises the particle geometry – the result is then introduced to the macroscale model via a convolution.

These different approaches lead to scenarios where each model may be more appropriate. For galvanostatic operations, the calculation of the convolution in DC^{1t} (or SC^{1t}) is straightforward,¹³ and is

therefore the more computationally efficient approach: there is no need to continually recompute solutions to the diffusion equation, as the DFN (or SPMe) does. Contrastingly, for potentiostatic operations, calculation of the convolution in DC^{1t} (or SC^{1t}) could be much more involved (see Ref. 13), and thus the DFN's approach may be preferable. To confirm this, further research would be required to explore the DC model's performance for such operations (only galvanostatic simulations were performed in the original publications).

Developing efficient ways to solve the time convolution within these models would be an interesting research perspective. For example, Fig. 5 shows that the closure variable's transient behavior is mainly important within a relatively small time window, and at later times $\partial_t \langle s^t \rangle_A = 0$. Therefore it may be possible to approximate $\partial_t \langle s^t \rangle_A \star j_k^{1t}$ at time t using a limited number of previous time steps, rather than performing the entire integral (Eq. 10). In the volume-averaging literature, alternatives to resolving the full time convolution can also be found in the form of phase-splitting or multi-rate mass-transfer models.^{30–32}

The general hybrid model.—Although not addressed until this point, an illustration of the general hybrid model (GHM) was included in Fig. 1. This is synonymous with the “homogenised model” reviewed by Brosa-Planella et al.¹ In this formulation, three out of the four conservation equations are upscaled/homogenised, exactly as in the DFN approach. However, the assumption that particles are isolated spheres is not made. Instead, a heterogeneous unit cell is used for the auxiliary particle domains. As in the DFN, all points $x \in \Omega$ are coupled to an auxiliary particle domain, Ω_k^x , but now the AM mass conservation PDE is solved on a (likely complex) 3D volume.

Performing an analysis analogous to that above, the GHM can be shown to be formally equivalent to DC^{1t} regardless of the electrode geometry. This equivalence only requires that the auxiliary particle domain geometry in the GHM is identical to that used to solve the closure problem for DC^{1t} (Eq. 11). This gives another insight into the DC approach. As highlighted by Brosa-Planella,¹ the GHM is extremely computationally expensive. The DC^{1t} model maintains the full geometrical complexity of the GHM, but shifts the computational burden into an ‘offline’ diffusion simulation (the closure problem), which is solved once to characterize the electrode. When solving the battery model, this information is then included via the time convolution.

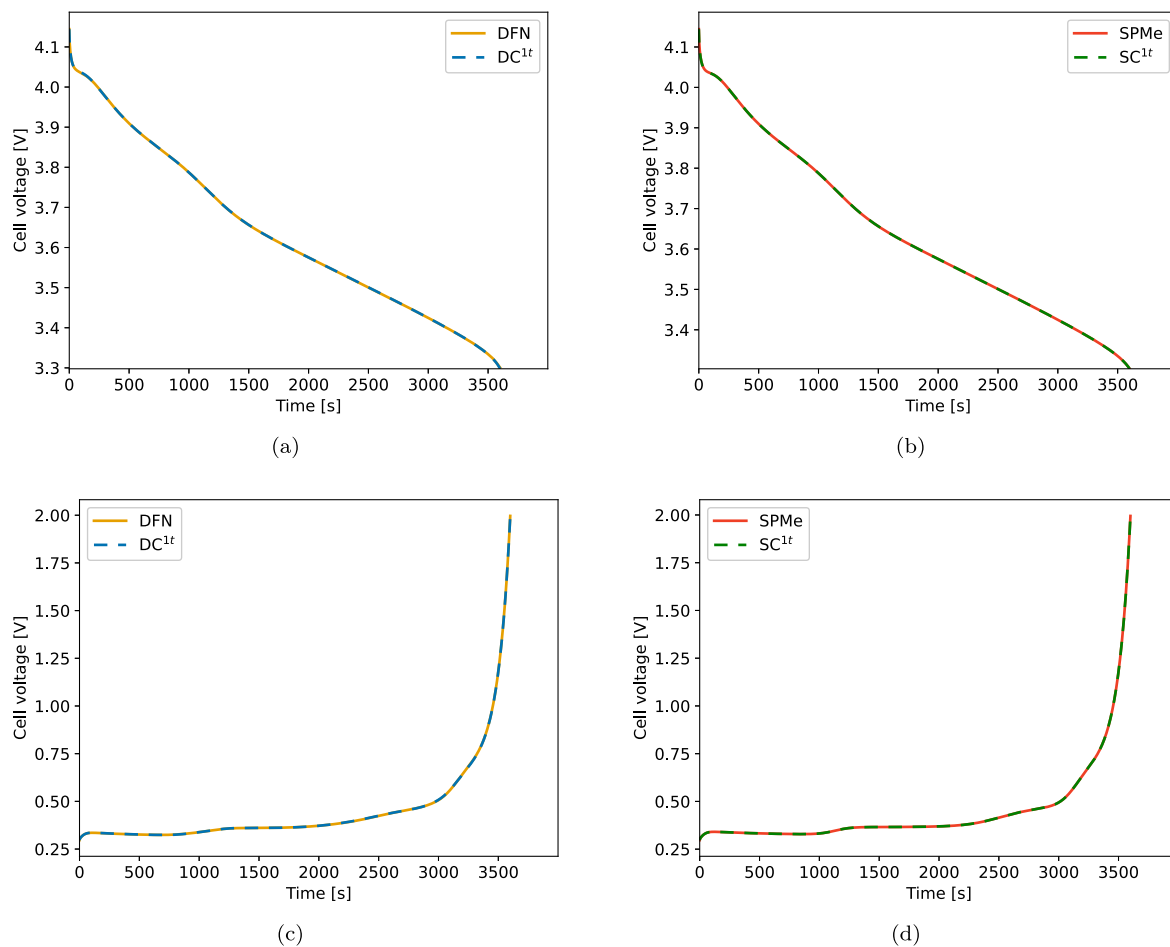


Figure 7. 1C discharge cell-voltage results for NMC (a), (b) and graphite (c), (d). DFN and DC^{1t} models are shown on the left (a), (c), with SPMe and SC^{1t} models on the right (b), (d).

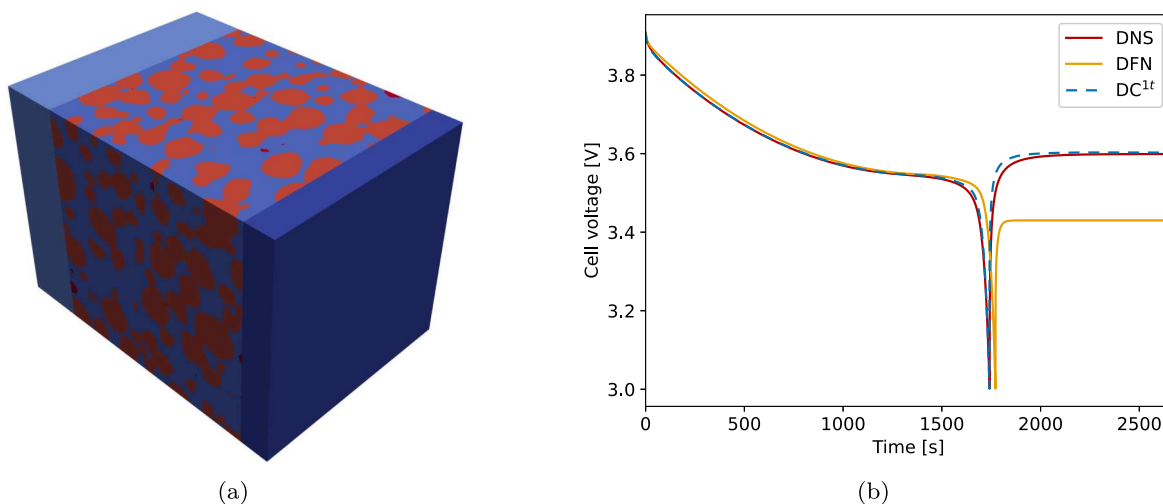


Figure 8. An NMC half cell reconstructed from a tomographic image (a) and the cell-voltage results for a discharge-relaxation simulation on this half cell (b), taken from our previous work.^{12,13} The DNS reference is the direct numerical solution to the microscale model, as described in the original publication.

Conclusions

In this work, we demonstrated that the Doyle–Fuller–Newman (DFN, a.k.a. P2D) and dual-continuum (DC) models of lithium-ion batteries are *formally* equivalent under certain conditions, which can be summarised as:

1. Electrode active-material particles are isolated spheres of uniform radius.
2. The transient formulation of the DC model is used.
3. The initial concentration is constant throughout the active-material phase.
4. The active-material diffusivity is not a function of radial position.

By analogy, we also show that single-particle models (SPM) are exactly equivalent to the single-continuum (SC) model under such conditions. The equivalence was first demonstrated mathematically, after which two half cells were simulated to confirm that model results were indeed identical.

This finding gives a fundamental insight into the nature, and advantages of the DC (or SC) model. Through its closure problem, the DC (or SC) approach can represent arbitrarily complex active-material microstructures, and reduces to the DFN (or SPM) for an idealised electrode of spherical particles. This equivalence also gives insight into the nature of the DFN—it can be interpreted as one approach to solving the time convolution describing memory effects in the diffusion process. We also discussed the equivalence between the DC model and a general hybrid model, which holds for any electrode geometry. Finally, two perspectives were proposed for future research: (i) validation of DC models for potentiostatic operations, and (ii) development of efficient methods to solve the time convolution within a battery modelling context.

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